

DATA- UNDERSTANDING

1. ADVERTISING

Name of Feature	Description	Type	Relevance
Daily_Time_Spent_on_Site	Average time (in minutes) spent by the user on the website	Quantitative, Continuous	Relevant – likely influences ad click behavior.
Age	Age of the user	Quantitative, Continuous	Relevant – user's age may impact likelihood to click on ads.
Area_Income	Annual income of the user's area	Quantitative, Continuous	Relevant – financial status may influence ad click behavior.
Daily_Internet_Usage	Average daily internet usage (minutes)	Quantitative, Continuous	Relevant – higher internet usage may correlate with ad engagement.
Ad_Topic_Line	Headline of the advertisement shown	Text, Categorical	Relevant – the content of the ad may affect clicks.
City	User's city	Categorical, Nominal	Potentially relevant – may capture geographic trends in behavior.
Male	Gender of the user (1 = Male, 0 = Female)	Categorical, Binary	Relevant – gender might influence ad engagement.
Country	User's country	Categorical, Nominal	Potentially relevant – geographic differences may affect clicks.
Timestamp	Date and time of the user's visit	Datetime	Potentially relevant – can extract day, month, time trends for analysis
Clicked_on_Ad	Whether the user clicked on the ad (target variable)	Categorical, Binary	Highly relevant – this is the target variable for prediction.

2 BANK DATA

<u>Name of Feature</u>	<u>Description</u>	<u>Type</u>	<u>Relevance</u>
default	Whether the customer has a default credit (0 = no, 1 = yes)	Categorical, Binary	Relevant – indicates financial reliability, may affect campaign outcome
balance	Customer's account balance	Quantitative, Continuous	Relevant – wealth may affect subscription likelihood
housing	Whether customer has a housing loan (0 = no, 1 = yes)	Categorical, Binary	Relevant – indicates financial commitments
loan	Whether customer has a personal loan (0 = no, 1 = yes)	Categorical, Binary	Relevant – may affect ability or interest to subscribe
duration	Duration of last contact in seconds	Quantitative, Continuous	Highly relevant – longer contact usually increases conversion probability
campaign	Number of contacts performed during this campaign	Quantitative, Discrete	Relevant – shows intensity of campaign
pdays	Days passed since last contact with customer (-1 means never contacted)	Quantitative, Continuous	Relevant – indicates recency of interaction
previous	Number of contacts performed before this campaign	Quantitative, Discrete	Relevant – prior interactions may influence outcome

3 COMPANY-DATA

<u>Name Of Feature</u>	<u>Description</u>	<u>Type</u>	<u>Relevance</u>
Sales	Sales value of the product (in thousands or units)	Quantitative, Continuous	Highly Relevant – This is likely the target variable
CompPrice	Price charged by competitors for the same product	Quantitative, Continuous	Relevant – May affect sales
Income	Average income of the target market	Quantitative, Continuous	Relevant – Higher income may increase sales
Advertising	Amount spent on advertising	Quantitative, Continuous	Relevant – Higher advertising spend can influence sales
Population	Population of the area	Quantitative, Continuous	Relevant – Larger population may result in higher sales
Price	Price of the product	Quantitative, Continuous	Relevant – Price sensitivity can affect sales
ShelveLoc	Quality of shelving location (Bad, Medium, Good)	Categorical, Ordinal	Relevant – Shelf placement affects visibility and sales
Age	Average age of customers in the area	Quantitative, Continuous	Relevant – Age affects buying behavior
Education	Average years of education of customers	Quantitative, Continuous	Relevant – Education may influence purchasing decisions

4 CRIME-DATA

<u>Name Of Feature</u>	<u>Description</u>	<u>Type</u>	<u>Relevance</u>
State	Name of the US State	Categorical, Nominal	Irrelevant for modeling numeric crime rates, but useful for labeling or grouping
Murder	Murder rate per 100,000 population	Quantitative, Continuous	Relevant – crime severity metric
Assault	Assault rate per 100,000 population	Quantitative, Continuous	Relevant – crime severity metric
UrbanPop	Percent of population living in urban areas	Quantitative, Continuous	Relevant – urbanization may influence crime rates
Rape	Rape rate per 100,000 population	Quantitative, Continuous	Relevant – crime severity metric

5 EAST-WEST-AIRLINES

<u>Name Of Feature</u>	<u>Description</u>	<u>Type</u>	<u>Relevance</u>
Id	Customer Identification Number	Quantitative, Nominal	Irrelevant – unique ID does not affect prediction
Balance	Account Balance	Quantitative, Continuous	Relevant – Customer's financial capacity may affect rewards eligibility
Qual_miles	Number Of Qualifying Miles	Quantitative, Continuous	Relevant – Indicates customer activity
Cc1_miles	Miles Earned From Credit Card 1	Quantitative, Continuous	Relevant – Contributes to total miles
Cc2_miles	Miles Earned From Credit Card 2	Quantitative, Continuous	Relevant – Contributes to total miles
Cc3_miles	Miles Earned From Credit Card 3	Quantitative, Continuous	Relevant – Contributes to total miles
Bonus_miles	Bonus Miles Earned	Quantitative, Continuous	Highly Relevant – Indicates reward accumulation
Bonus_trans	Number Of Bonus Transactions	Quantitative, Continuous	Relevant – Engagement metric
Flight_miles_12mo	Flight Miles In Last 12 Months	Quantitative, Continuous	Highly Relevant – Shows recent travel activity

6 HEART DISEASE

<u>Name Of Feature</u>	<u>Description</u>	<u>Type</u>	<u>Relevance</u>
Age	Age of the patient in years	Quantitative, Continuous	Relevant – Age influences heart disease risk
Sex	Sex of the patient (1 = Male, 0 = Female)	Categorical, Binary	Relevant – Gender influences heart disease risk
Cp	Chest pain type (0–3)	Categorical, Ordinal	Relevant – Chest pain type correlates with heart disease severity
Trestbps	Resting blood pressure (mm Hg)	Quantitative, Continuous	Relevant – Blood pressure is a risk factor
Chol	Serum cholesterol (mg/dl)	Quantitative, Continuous	Relevant – Cholesterol level affects heart disease risk
Fbs	Fasting blood sugar > 120 mg/dl (1 = True, 0 = False)	Categorical, Binary	Relevant – High blood sugar indicates risk
Restecg	Resting electrocardiographic results (0–2)	Categorical, Ordinal	Relevant – ECG results indicate heart health
Thalach	Maximum heart rate achieved	Quantitative, Continuous	Relevant – Heart rate response is important
Exang	Exercise-induced angina (1 = Yes, 0 = No)	Categorical, Binary	Relevant – Indicates heart strain
Target	Presence of heart disease (1 = Yes, 0 = No)	Categorical, Binary	Highly Relevant – Target variable to predict

7 MTCARS

<u>Name Of Feature</u>	<u>Description</u>	<u>Type</u>	<u>Relevance</u>
Mpg	Miles per gallon (fuel efficiency)	Quantitative, Continuous	Highly Relevant – Key performance metric
Cyl	Number of cylinders	Quantitative, Discrete	Relevant – Engine configuration affects performance
Disp	Engine displacement (cubic inches)	Quantitative, Continuous	Relevant – Indicates engine size and power
Hp	Gross horsepower	Quantitative, Continuous	Relevant – Engine power affects performance
Drat	Rear axle ratio	Quantitative, Continuous	Relevant – Influences mechanical performance
Wt	Weight of the car (1000 lbs)	Quantitative, Continuous	Relevant – Heavier cars have lower performance
Qsec	1/4 mile time (seconds)	Quantitative, Continuous	Relevant – Acceleration performance
Vs	Engine type (0 = V-shaped, 1 = Straight)	Categorical, Binary	Relevant – Engine type affects performance
Am	Transmission (0 = Automatic, 1 = Manual)	Categorical, Binary	Relevant – Transmission type affects driving performance
Gear	Number of forward gears	Quantitative, Discrete	Relevant – Influences acceleration and speed
Carb	Number of carburetors	Quantitative, Discrete	Relevant – Engine fuel intake configuration